Rajarata University of Sri Lanka
Department of Computing

ICT1402 PRINCIPLES OF PROGRAM DESIGN & PROGRAMMING

Lecture 03

Flow Charts and Design Structures

K.A.S.H. Kulathilake
Ph.D., M. Phil., MCS, B.Sc. (Hons) IT, SEDA(UK)

Objectives

- At the end of this lecture students should be able to;
  - Define the major flow chart design symbols.
  - Select appropriate flow chart design symbol for a design statement.
  - Define three basic design structures.
  - Combine flow chart design symbols into the basic structures.
  - Define the nested structures in flow charts.
  - Differentiate two major types of loop construct.
  - Design complete flow charts for the real-world requirements.

Introduction

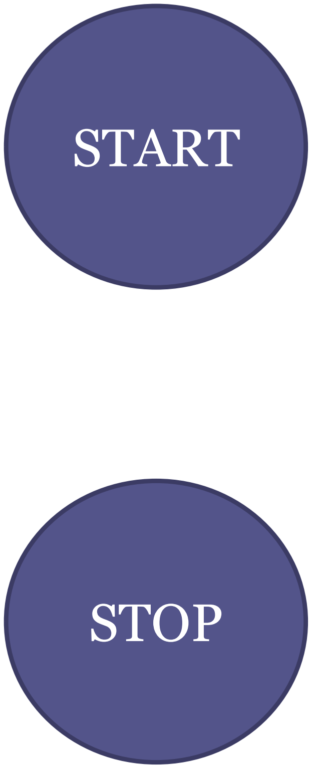
- Flow chart is a symbol-oriented design system that identifies the type of statement by the shape of the symbol containing the statement.
- Once we have created our list of tasks and sub tasks, it is required to convert those tasks to symbols in the flowchart.
- Generally, the lowest level of our subtasks will each become one symbol in the flow chart.
- These symbols are connected by flow lines that show the flow of control through the program.

Flow Chart Symbols

- Flow Lines
  - Representing using arrow.
  - It shows the flow control through the program.

Flow Chart Symbols (Cont…)

- Terminal Symbol
  - It is used to indicate beginning and end of a subsection within the flowchart, known as a module.
  - The terminal symbol will have only one flow line since a Start will have an exit flow line but no entering flow line and a Stop will have an entering flow line, but no exit.



Flow Chart Symbols (Cont…)

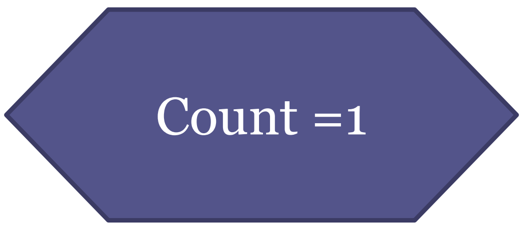
- Input Output Symbol
  - It is used to indicate both input(READ) and output(WRITE).
  - Be specific when indicating what is Reading or Writing
    - e.g. Do not say Read Recode say Read Name, Address
  - Input means the program is receiving data from an external source, something connected to the computer but not part of the main processing/ memory core.
    - e.g. keyboard, a disk drive, a tape, file
  - The received values are stored in memory.
  - Output indicates the program is sending data from memory to an external device like a printer, a monitor screen, or a disk drive.

Flow Chart Symbols (Cont…)

- Process Symbol
  - Indicates using a rectangle and represents calculations and initialization of memory locations.
  - An initialization means to assign a starting value to a location in memory, like count = 0.
  - A calculation means to perform some calculation and assign the result of the calculation to a location in memory.
  - The calculation could be as simple as ADD 1 to count, or very complex.

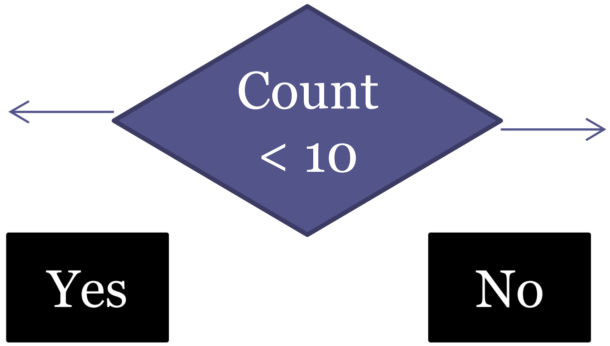
Flow Chart Symbols (Cont…)

- Preparation Symbol
  - This symbol is used for any initialization done, usually at the beginning of a program.



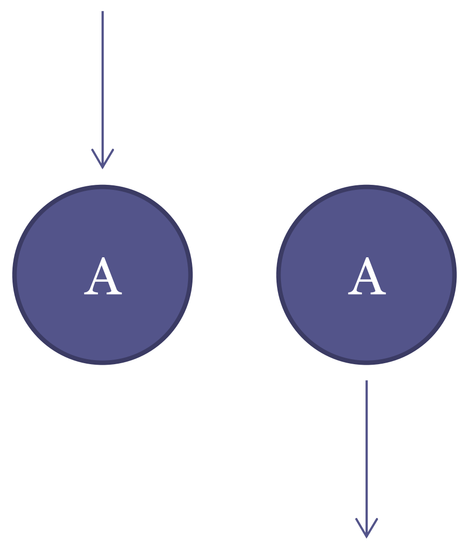
Flow Chart Symbols (Cont…)

- Decision Symbol
  - The diamond either asks a YES/NO questions OR makes a True/ False statement.
  - There will always be two exits from a decision symbol, one labeled YES or TRUE and the other labeled NO or FALSE.
  - This is the only symbol that can have two exits.



Flow Chart Symbols (Cont…)

- Connector
  - This is used to connect the flow chats to represent its' continuation

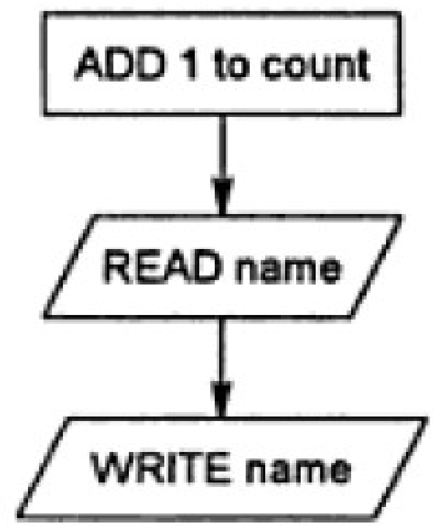


Basic Design Structures

- There are three generally accepted design structures;
  - Sequence
  - Selection
  - Iteration/ Loop
- You should limit your designs to these structures.
- Every structure has a single entrance flow line and a single exit flow line.

Sequence Structure

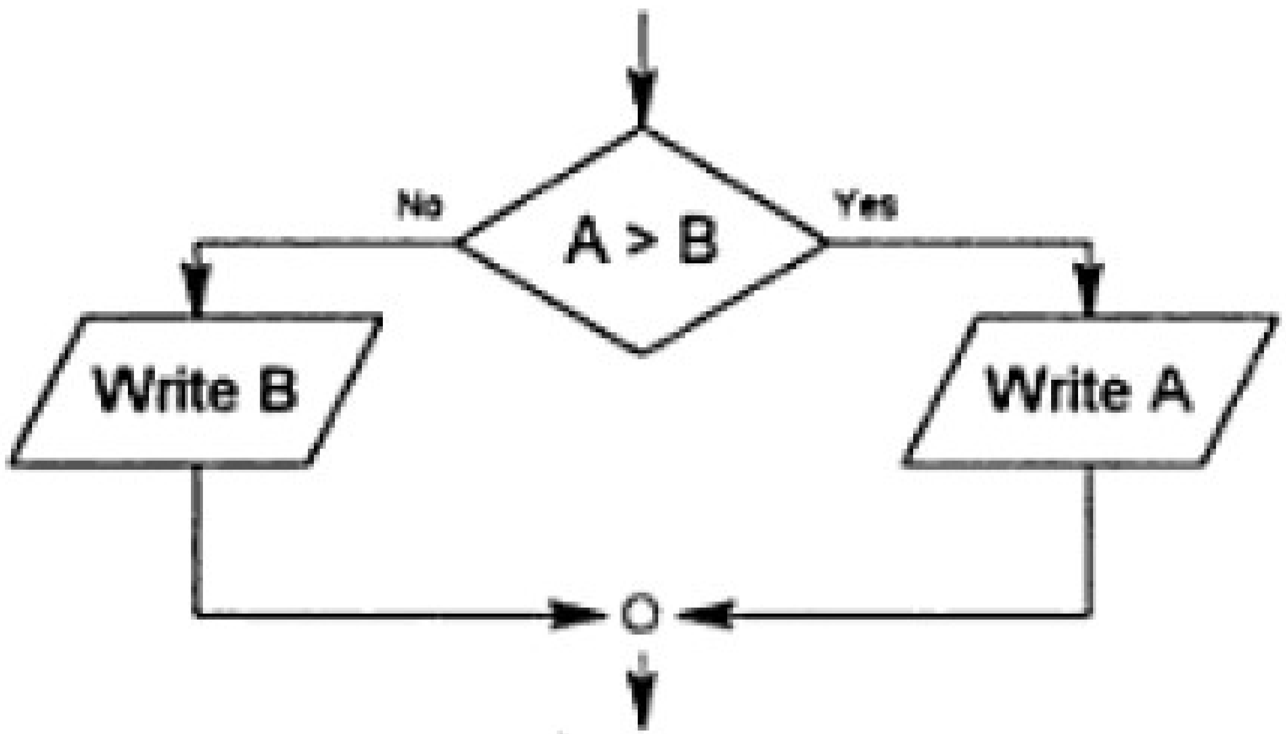
- It is the simplest because it is nothing more than a series of statements performed one after another.
- It is a sequence because the flow lines always continue in the same direction when the structure is executed.
- A sequence could include many instructions, but each would be done in turn, in the same order.



Selection Structure

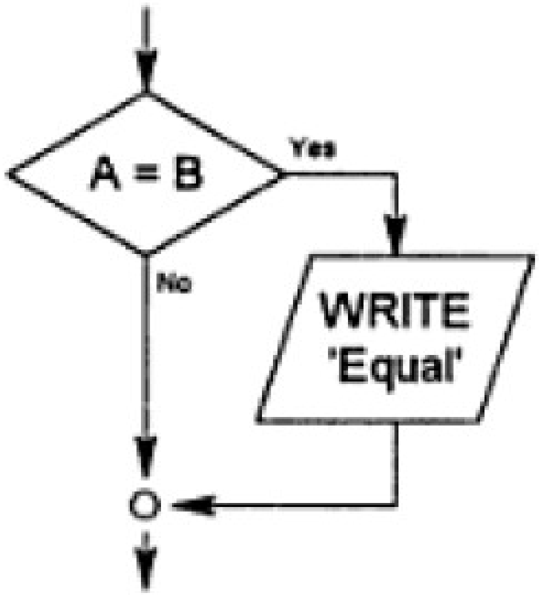
- It uses the decision symbol to ask a question.
- The structure includes the decision plus the operations done in response to the decision.
- There will be two exits from the decision box, but both branches are still a part of the section structure.
- Further, both branches must rejoin a single flow line to exit the structure.

Selection Structure (Cont…)



Selection Structure (Cont...)

- One Sided Selection
  - If the selection only has an operation on the true side (or less commonly, only on the false side), it is called a one-sided selection.
  - With the one-sided structure, the "null" or empty side still must have its exit from the decision box, but there are no operations along the flow line.



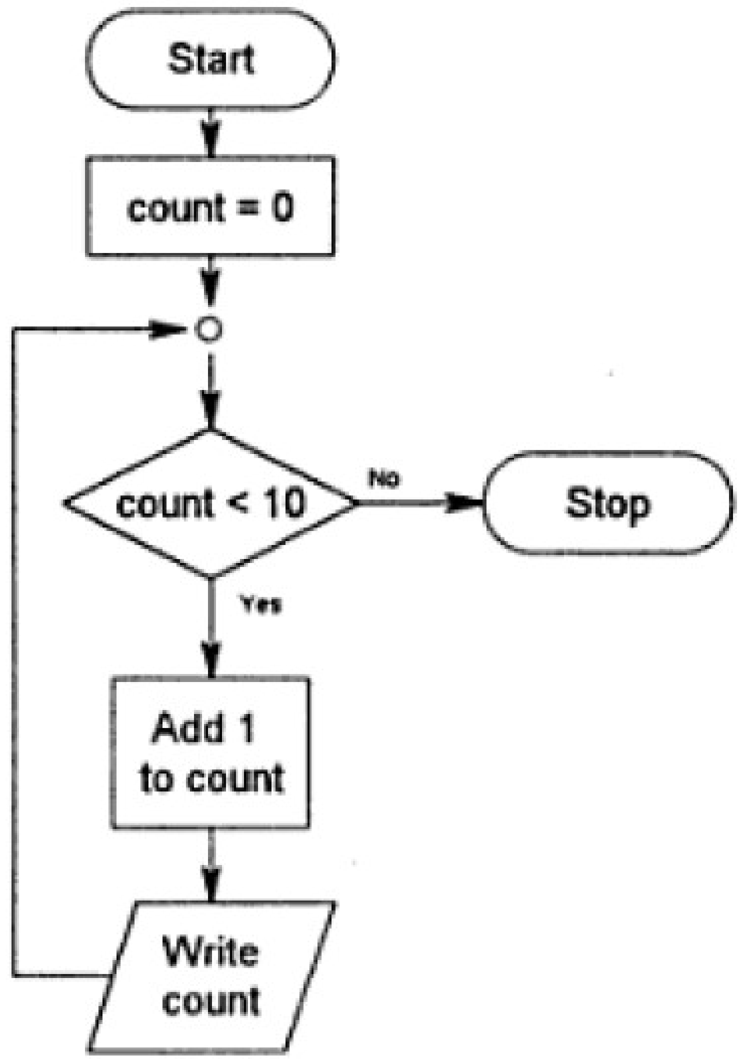
Loop Structure

- It also uses the decision symbol since there must always be a way to get out of the loop.
- The loop structure in general will have a decision, one or more additional symbols indicating the steps to be done within the loop, and a flow line that sends control back to the beginning of the loop.
- There are two main loop types:
    - The while loop and
    - the Until loop.

Loop Structure

- While Loops
  - While loops are leading decision loops, meaning that the decision symbol always is the first statement of the loop.
  - If the test is true, control follows the flow line that goes into the loop.
  - If the test is false, control follows the flow line to the first statement after the loop.
  - In other words, we stay in the loop WHILE the decision is true.

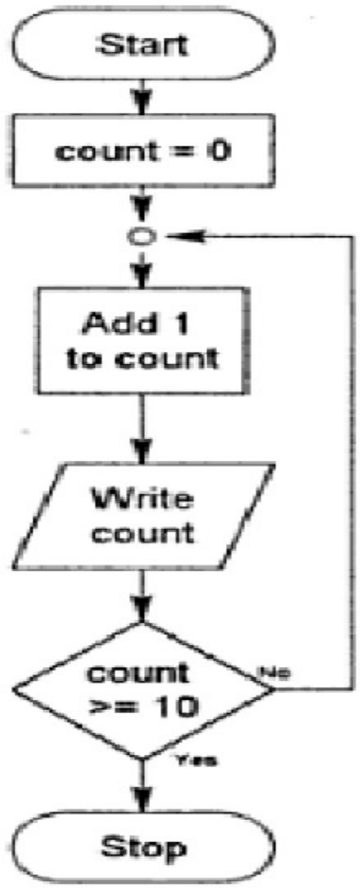
Loop Structure (Cont…)



Loop Structure (Cont…)

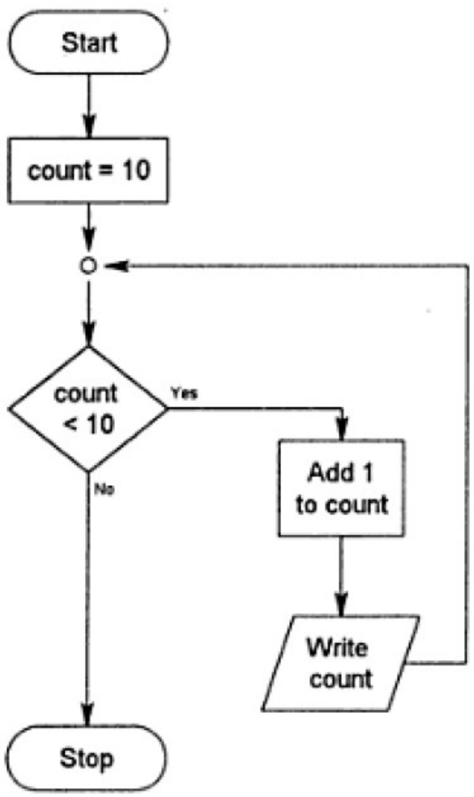
- Until Loop
  - Until loops are trailing decision loops, meaning the decision is always the last statement of the loop.
  - The body of the loop will be executed once before control even gets to the loop test.
  - In the until loop, if the test is false, control follows the flow line back to repeat the loop.
  - If the test is true, control follows the branch from the decision box that exits the loop structure.
  - We continue to repeat the loop UNTIL the decision is true.

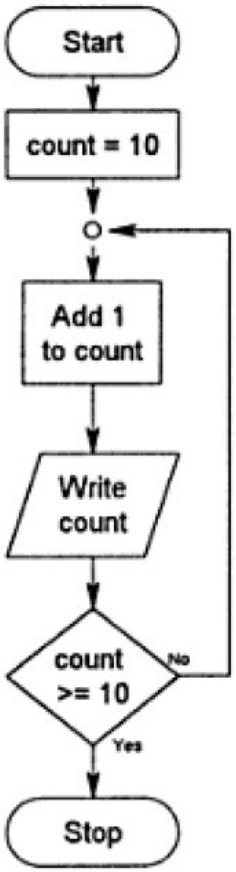
Loop Structure (Cont…)



Loop Structure (Cont…)

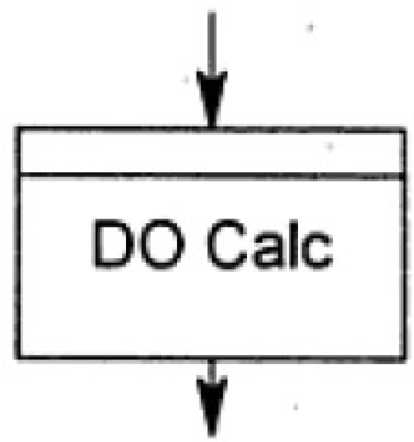
- Difference between two loop types





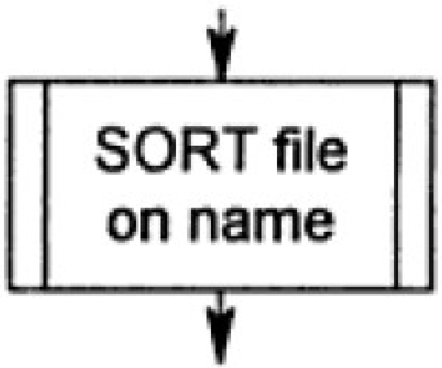
Internal Process

- This variation of the process symbol indicates a call to a module that is within the same program.
- Internal process symbols are used to represent sub-modules.
- A "call" to an internal process transfers control to that process(module) without using flow lines to connect the two parts.
- An internal process helps us keep our design (our resulting program) organized.
- We can group tasks that are related into one module, so they can be easily checked and modified.



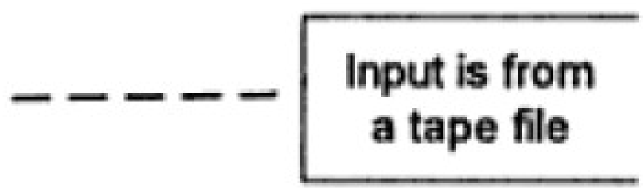
External Process

- Another variation of the process symbol, this symbol indicates a call to a module that is not included within the current program.
- An external module would be something like a “utility” made available by the operating system or compiler or some other program.
- The external process call also indicates an automatic branch and return, but this time the branch is to a set of instructions outside the program.



Annotation Symbol

- Allows you to add comments to your design without cluttering the flowchart itself.
- The comment is connected to the relevant box or area with a dashed line (so it isn't confused with a flow line).



Example: Payroll Example

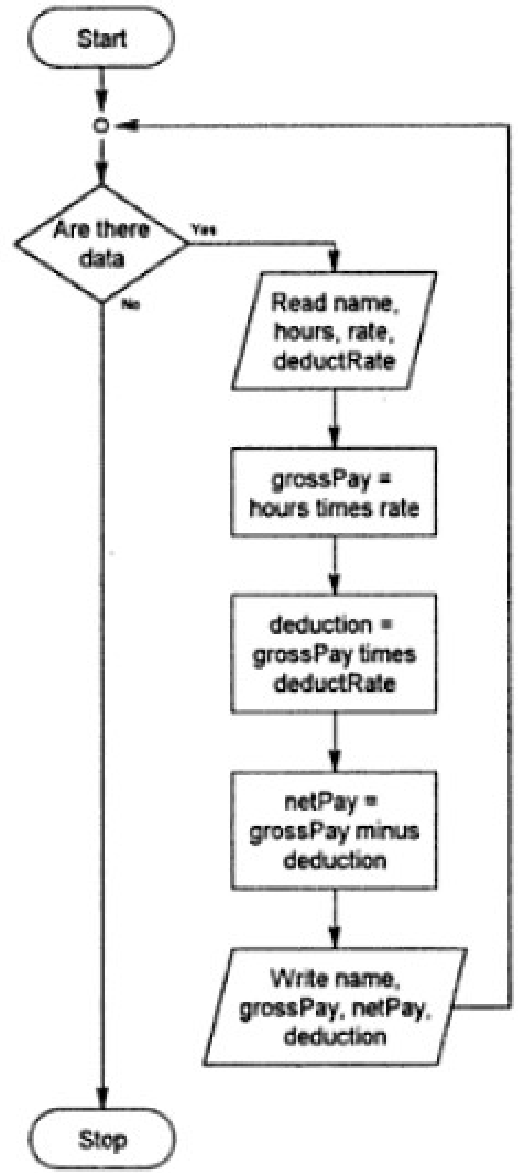
- Task list

Read record
Calculate values
Write output

<u>Calculate values subtasks</u>
Gross pay = hours times pay rate
Deductions = gross pay times deduction rate
Net pay = gross pay minus deductions

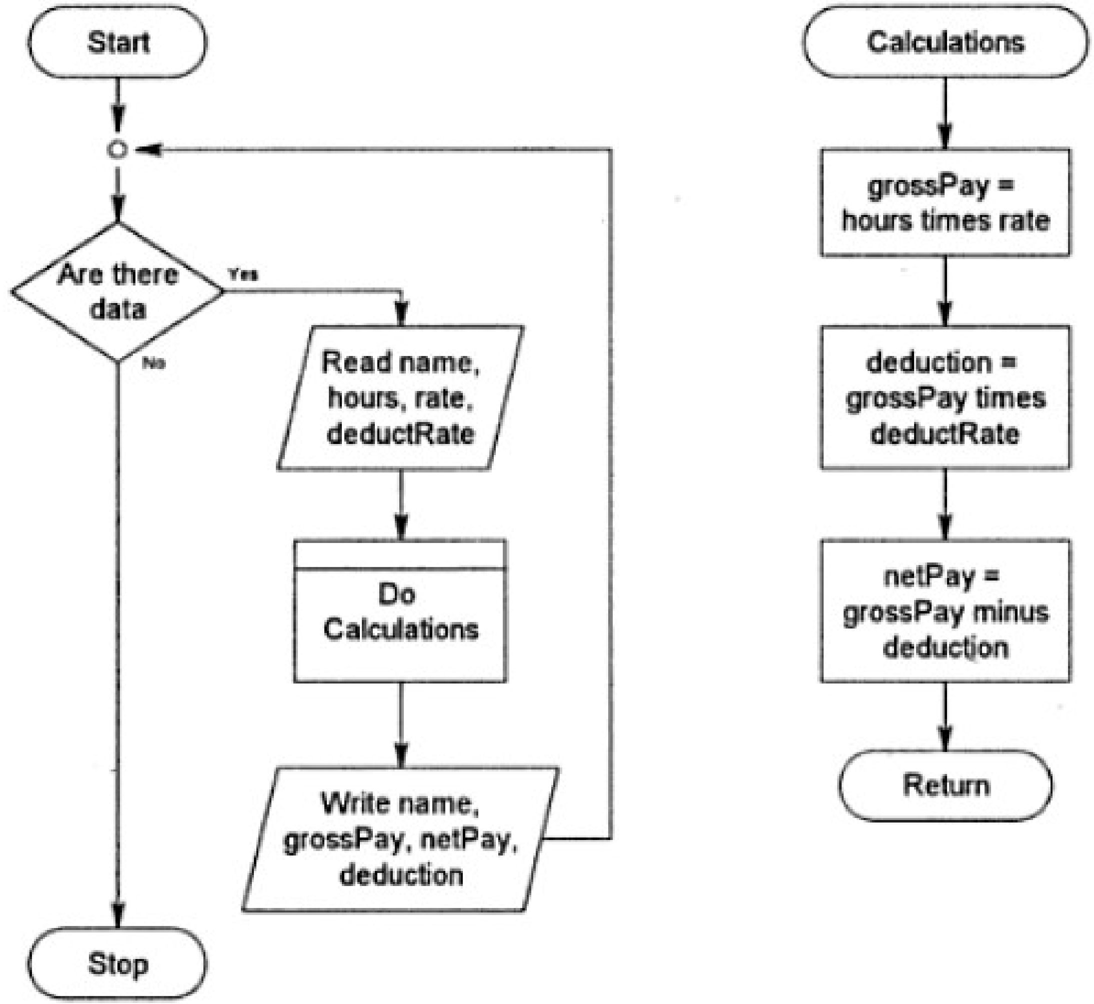
Example: Payroll Example

- Now let's put them into a flowchart.
- The chart that follows includes a while loop with a sequence inside a loop.



Example: Payroll Example

- Move the calculations into a module.
- The idea is to take the details and "force" them into a lower level in the design.
- The "call/ internal process symbol" transfers the control to the module.
- At the end of the module execution (RETURN), it transfers the control back to the statement that follows the call.



Problems: Identifying Modules

- You have the list of names and GPA for all the students graduating at the end of this semester. You want to create a report of the names of all students who should be included in the Dean's List. To be eligible for the Dean's list, a students must have a GPA of at least 3.5.
- You have a file containing the widths and heights for multiple rooms. You want to calculate the number of gallons of paint needed to paint the walls in each room. Do not worry about subtracting out for doors and windows. Assume that one gallon of paint will cover 500 square feet. Write out the width, height and number of gallons. If 5 or more gallons will be needed for one room, also write out the message " Look for quantity discount".

Program Control

- One of the most important aspects of programming is controlling which statement will execute next.
- *Control structures* / *Control statements* enable a programmer to determine the order in which program statements are executed.
- These control structures allow you to do two things:
    - skip some statements while executing others, and
    - repeat one or more statements while some condition is true.

Program Control (Cont…)

- Can observe in
  - Sequence
  - Selection
  - Iterations/ loop
  - Calling a internal or external module

Basic Input/ Output

- Input
  - Reading internal or external data which is required to initiate the program.
    - Raw data (interactive data).
    - File read.
    - Reading a data store.
    - Return values from internal or external module.
- Output
  - Simply Results
    - Display on screen
    - Pass/ return to another module
    - State change
    - Control change

Objectives - Revisit

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- Differentiate two major types of loop construct.
- Design complete flow charts for the real world requirements.

References

- Chapter 02 of Computer Program Design, Elizabeth A. Dickson.

Q & A

Next: Pseudocode